Erratum: "Shift-symmetric configurations in two-dimensional cellular automata: Irreversibility, insolvability, and enumeration" [Chaos 29, 063120 (2019)]

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In the original article,[1] Eqs. (32) and (33) contain a misprint, the range stated for the k-active-cell formulas is too narrow, the standing assumptions are left implicit, and the proofs of Lemmas E.21, A.21, and A.22 have gaps. We correct these points below. Equation and lemma numbers refer to Ref. 1. As there, $n=\prod_{i=1}^{k} p_i^{\alpha_i}$ with $k=\omega(n)$; for $\mathbf{v}\in\{0,1,2\}^m$ we write $|\mathbf{v}|=v_1+\cdots+v_m$, and "$\mathbf{0}\lhd\mathbf{v}\trianglelefteq\mathbf{2}$" means $\mathbf{v}\neq\mathbf{0}$.

1. *Standing assumptions.* Ref. 1 states its results without restricting n and $|\Sigma|$. They require $n\geq 2$ and $|\Sigma|\geq 2$ (the proofs use $|\Sigma|\geq 2$) and, for k active cells, $0\leq k\leq n^2$. Outside this range some statements fail for trivial reasons. For $n=1$ there is no nonzero translation, so $|S_{1\times 1}|=0$ and the lower bounds in Eqs. (21), (23), and (24) fail. For $|\Sigma|=1$ the only configuration is shift-symmetric; both sides of Eq. (21) then equal 1, so "only if n is a prime" fails at every composite n, and the lower bound in Eq. (24) reads $n\leq 1$. Equation (28) holds for $|\Sigma|\geq 2$ and $0\leq k\leq n^2$.

2. *Eqs. (32) and (33) (Theorem E.32 and Corollary A.33).* The factors $r(i)$ must be built from the primes r_i of $d=\gcd(k,n)$, not from the primes p_i of n. The correct definition is
$$r(i)=\begin{cases}1 & \text{if } v_i=0,\\ r_i+1 & \text{if } v_i=1,\\ r_i & \text{if } v_i=2.\end{cases}$$
As printed (primes in increasing order), Eqs. (32) and (33) give 36 for $n=6$, $k=3$, $|\Sigma|=2$, whereas $|S^{a}_{6\times 6,3}|=48$: each of the four subgroups of order 3 fixes $\binom{12}{1}=12$ configurations with three cells in state a, and no configuration is fixed by two of them. Eqs. (30) and (31) give 48; they are correct as printed. The printed and the corrected formulas agree when n is a prime or a prime power, and in general whenever $r_i=p_i$ for every $i\leq\omega(d)$.

3. *Range of k and a convention in Eqs. (30)–(33).* In Lemmas E.30 and E.31 and Theorem E.32, the hypothesis "$k\leq n$" should read "$0\leq k\leq n^2$", the range over which Eq. (34) sums; Eq. (33) holds on the same range. In Eqs. (30)–(33), $\binom{n^2/h}{k/h}$ (with $h=h(\mathbf{u})$ or $h(\mathbf{v})$) is zero when k/h is not an integer, and a vanishing binomial coefficient annihilates its whole summand, including the factor $(|\Sigma|-1)^{(n^2-k)/h}$ with a fractional exponent. For $|\Sigma|\geq 2$, Eq. (29) needs no such convention.

4. *Proof of Lemma E.21.* The proof begins "If $k=1$, i.e., n is a prime, the equality holds as shown in Corollary E.20." But $k=\omega(n)=1$ also for prime powers $n=p^{\alpha}$, $\alpha\geq 2$, where equality fails: for $n=4$ and $|\Sigma|=2$, $|S_{4\times 4}|=736$, while the left side of Eq. (21) is 72. The case split should read "n prime" and "n composite". For every composite n, including prime powers, $p_1\leq n/2$ and $|\Sigma|^n\geq 2^n\geq n+1$ give
$$\begin{aligned}|S_{n\times n}|\geq q^1_{n\times n}&=|\Sigma|^{n^2/p_1}(p_1{+}1)-|\Sigma|^{n^2/p_1^2}p_1\geq|\Sigma|^{n^2/p_1}\\&\geq|\Sigma|^{2n}\geq(n{+}1)|\Sigma|^n>(n{+}1)|\Sigma|^n-n|\Sigma|,\end{aligned}$$
where $|S_{n\times n}|=q^k_{n\times n}\geq\cdots\geq q^1_{n\times n}$ by Lemma A.21 (see item 5). This also justifies the final strict inequality of the published proof, which is stated there without justification. Lemma E.21, including "equality if and only if n is a prime", therefore holds as stated.

5. *Proofs of Lemmas A.21 and A.22.* Both proofs use the decomposition $q^{m+1}_{n\times n}=x_0+x_1+x_2+y_1+y_2$ of Ref. 1, with $p=p_{m+1}$, $A=n^2/p$, $x_1=-(p{+}1)\,q^m_{n\times n}(p^{-1})$, $x_2=p\,q^m_{n\times n}(p^{-2})$, $y_1=(p{+}1)|\Sigma|^A$, and $y_2=-p|\Sigma|^{A/p}$. The two inequality chains of these proofs have three defects. (a) The last step of the chain in the proof of Lemma A.21 needs $A/2\geq 4\log_2 n+1$, which is false for $(n,m)=(6,1),(10,1),(14,1)$; the last step of the chain for Lemma A.22 needs $\frac{5}{6}A\geq 4\log_2 n$, which is false for $(n,m)=(6,1)$ (primes in increasing order; for any other order the failures occur only at these n). (b) The first inequality of the chain for Lemma A.22 is wrong as printed: its right side drops the factors $p/(p{+}1)$ of $q^m_{n\times n}(p^{-2})$ and of $|\Sigma|^{A/p}$, which makes it larger than the left side by $(|\Sigma|^{A/p}-q^m_{n\times n}(p^{-2}))/(p{+}1)$. This is positive for every n, because by Lemma R1 $q^m_{n\times n}(p^{-2})$ counts some, but not all, of the $|\Sigma|^{A/p}$ configurations on $\mathbb{Z}^2_{n/p}$; for $n=6$ and $|\Sigma|=2$ the two sides are 4278 and 4280. (c) The proofs use without justification that $x_2\geq 0$ (Lemma A.21) and that $q^m_{n\times n}(p^{-1})\geq 0$ and $q^m_{n\times n}(p^{-2})\geq 0$ (Lemma A.22); Lemma R1 proves both.

The following three lemmas replace both chains entirely; they hold for every order of the primes. With them, Lemmas A.21 and A.22, and hence E.21 and E.22, hold as stated.

Lemma R1 *Let $T = \mathbb{Z}_a \times \mathbb{Z}_b$ with $p_i \mid \gcd(a,b)$ for $i \le m$, and let L_i be the set of subgroups of T of order p_i. For $H \le T$ let $\mathrm{Fix}(H)$ be the set of configurations on T that are constant on the cosets of H. Then*

$$\Big|\bigcup_{i\le m}\bigcup_{C\in L_i}\mathrm{Fix}(C)\Big| = \sum_{\mathbf{0}\lhd\mathbf{v}\trianglelefteq\mathbf{2}} (-1)^{1+|\mathbf{v}|}\,|\Sigma|^{ab/\prod_i p_i^{v_i}} \prod_{i=1}^{m} r(i),$$

with $r(i) = 1, p_i{+}1, p_i$ for $v_i = 0,1,2$. For $c = p^{-1}$ ($T = \mathbb{Z}_n \times \mathbb{Z}_{n/p}$) and $c = p^{-2}$ ($T = \mathbb{Z}_{n/p} \times \mathbb{Z}_{n/p}$) the right side is $q^m_{n\times n}(c)$, and

$$0 \le q^m_{n\times n}(c) \le \sum_{i\le m}(p_i{+}1)|\Sigma|^{n^2c/p_i} \le \frac{2n}{p}\,|\Sigma|^{n^2c/2}. \qquad (*)$$

Proof. Every $H \le T$ acts freely on T by translation, so $|\mathrm{Fix}(H)| = |\Sigma|^{ab/|H|}$, and $\mathrm{Fix}(C_1)\cap\cdots\cap\mathrm{Fix}(C_j) = \mathrm{Fix}(\langle C_1,\dots,C_j\rangle)$. Since $p_i \mid \gcd(a,b)$, the p_i-torsion of T is $\mathbb{Z}_{p_i}^2$; hence $|L_i| = p_i + 1$, two members of L_i generate a subgroup of order p_i^2, and subgroups for distinct primes generate their direct product. So $|\langle\bigcup_i J_i\rangle| = \prod_i p_i^{\min(|J_i|,2)}$ for $J_i \subseteq L_i$. Inclusion–exclusion over the nonempty $J = \bigcup_i J_i$, grouped by $v_i = \min(|J_i|,2)$, gives the identity, because $\sum_{j\ge2}(-1)^j\binom{p_i+1}{j} = p_i$. For $c = p^{-1}$ and $c = p^{-2}$ we have $p_i \mid n/p$ for $i \le m$ and $ab = n^2c$. The bounds in $(*)$ are the union bound and $\sum_{i\le m}(p_i{+}1) \le 2\sum_{i\le m} p_i \le 2\prod_{i\le m} p_i \le 2n/p$. □

Lemma R2 (replaces the chain for A.21) *If $\omega(n) \ge 2$, $1 \le m < \omega(n)$, and $|\Sigma| \ge 2$, then $y_1 + x_1 + y_2 \ge 0$. Hence $q^{m+1}_{n\times n} = x_0 + (y_1{+}x_1{+}y_2) + x_2 \ge x_0 = q^m_{n\times n}$, using $x_2 \ge 0$ from $(*)$.*

Proof. As $p_1\cdots p_m \ge 2$ divides n/p, we have $p \le n/2$ and $A \ge 2n$. By $(*)$ with $c = p^{-1}$, and since $p|\Sigma|^{A/p} \le (p{+}1)|\Sigma|^{A/2}$,

$$y_1 + x_1 + y_2 \ge (p{+}1)\,|\Sigma|^{A/2}\Big(|\Sigma|^{A/2} - \frac{2n}{p} - 1\Big) \ge 0,$$

because $|\Sigma|^{A/2} \ge 2^n \ge n+1 \ge 2n/p+1$. □

Lemma R3 (replaces the chain for A.22) *In the same setting, $x_1 + x_2 + y_2 \le y_1$; hence $q^{m+1}_{n\times n} \le q^m_{n\times n} + 2(p{+}1)|\Sigma|^{n^2/p}$, the recursion of the proof of Lemma A.22.*

Proof. By $(*)$, $x_1 \le 0$, and $y_2 \le 0$. By $(*)$ with $c = p^{-2}$, $x_2 \le 2n|\Sigma|^{n^2/(2p^2)}$. Since $n^2/p - n^2/(2p^2) \ge \frac{3}{4}A \ge \frac{3}{2}n$ and $2n \le 2^{3n/2}$, this gives $x_2 \le (p{+}1)|\Sigma|^{n^2/p} = y_1$. □

6. *Lemma A.23.* Corollary A.24 applies Lemma A.23 with p replaced by $p-1, p-2, \dots, 3$, which need not be prime divisors of n. The proof of Lemma A.23 holds verbatim for every integer $2 \le p \le n$, so Corollary A.24 and Lemma E.22 stand.

Effect on the article. No figure and no numerical value of Ref. 1 changes. Figures 6–8 and the values of $P^{\mathrm{unif}}_{n\times n}$ depend only on Eq. (19), which is correct. The density-uniform probabilities agree with the corrected Eq. (33): for $|\Sigma| = 2$, $P^{\mathrm{dens}}_{45\times45} = 9.87\times10^{-4}$ ("0.001" in Sec. IV A) and $P^{\mathrm{dens}}_{100\times100} = 2.000\times10^{-4}$ ("2×10^{-4}" in Sec. VI), and the plotted curve of Fig. 9 agrees with the corrected formula to within 0.1%. With items 1–6, Eqs. (21)–(24) and (28)–(33) and Lemmas E.21, E.22, and A.21–A.24 hold as stated, with $r(i)$ in Eqs. (32)–(33) as corrected in item 2. These corrections do not affect the conclusions of the article.

All counts and inequalities quoted here were verified in exact integer arithmetic, including against exhaustive enumeration, and the corrected Eqs. (30)–(33) and the statements of Lemmas E.21 and E.22 are machine-checked in Lean 4; scripts and proofs are in the repository of Ref. 2.

References

[1] P. Banda, J. Caughman, M. Cenek, and C. Teuscher, "Shift-symmetric configurations in two-dimensional cellular automata: Irreversibility, insolvability, and enumeration," Chaos **29**, 063120 (2019). https://doi.org/10.1063/1.5089889

[2] P. Banda, J. Caughman, and T. Lehman, "Shift-symmetric configurations in d-dimensional cellular automata: Enumeration, bounds, irreversibility, and detection" (2026), repository https://github.com/peterbanda/d-dim-shift-symmetry.